MASTER'S THESIS

Thesis submitted in fulfillment of the requirements for the degree of Master of Science in Engineering at the University of Applied Sciences Technikum Wien - Degree Program Data Science

Interpretable energy load forecasting in renewable energy dominated grids

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Wien, April 10, 2026

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Schlagworte: Schlagwort1, Schlagwort2, Schlagwort3, Schlagwort4

Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Keywords: Keyword1, Keyword2, Keyword3, Keyword4

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1 Introduction

1.1 Motivation of the topic

The set goals to drastically reduce greenhouse gas emissions to net-zero C02 are considered one of the greatest challenges for humanity in the upcoming decades. As a consequence, the existing energy infrastructure must undergo a deep transformation not seen before during this decarbonization phase. The resulting increase in distributed energy resources (DER) like electric vehicles and heat pumps as well as photovoltaics will evidently change existing load and power generation patterns used for electric load forecasting today.

The field of energy load forecasting (ELF) is under active research due to its crucial role in modern power systems [1]. Accurate load forecasts support a wide range of applications including demand-response optimization, energy trading, power system scheduling, maintenance planning and capacity sizing for various stakeholders such as transmission system operators (TSO), distribution system operators (DSO), industrial consumers and building energy managers [1], [2], [3], [4]. TSOs are responsible for maintaining a reliable grid and are challenged by the prospect of the changing electrical load and power generation patterns [4], [5].

Easier to understand statistical models have been historically the go to approach for ELF including linear regression models as well as time series models such as ARIMA and its seasonal extension SARIMA [5], [6]. These assume that electrical load can be determined by a linear combination of the previous load values and other exogenous variables [2]. More recently, decomposition-based models such as Prophet have gained popularity as they offer flexible modelling of seasonality, trends and holiday effects while remaining interpretable [7].

Interpretability is often a vaguely used term in the literature but Baur et al. defines it as the ability of humans to understand the models decision process just by understanding the models design [1]. Neural networks are nowadays gaining more attention in research because they can work with large amounts of data, have good performance and can handle complex relations of variables. Commonly used architectures in the literature include Long Short-Term Memory networks (LSTMs), XGBoost, one-dimensional Convolutional Neural Networks (1D CNNs), Support Vector Regression (SVR), as well as various ensemble and hybrid approaches that combine these methods [2], [3], [4], [7], [8]. Yet these substantially more complex approaches do not offer insights into the model's intrinsic processes, leaving them opaque. This prompts a surge in explainability, defined as the ability to reason about the factors that influenced a model's prediction [1]. Currently a wide variety of model specific and agnostic explainability techniques for artificial intelligence exist (XAI). A notable work from Zheng et al. implements a such post-hoc XAI methods, like SHAP, for ELF which illustrates that these methods can be

used to enhance the explainability of more complex models [9]. However, the implementation does not include confidence intervals, leaving operators without a clear sense of prediction risk. This represents a problem that often gets mentioned within literature, which is the lack of a quantified uncertainty within the point estimates from interpretable models [1], [10]. TSOs must reliably assess risks and when to plan reserves. Traditional point forecasts provide a single expected load value but no information on the uncertainty or risk around that prediction. This is a reason why the domain of probabilistic load forecasting (PLF) receives growing recognition, where the models not only provide point forecasts but also depict the uncertainty in their estimates over prediction intervals or scenarios.

Generally, Baur et al. describe how the different types of PLF models can be assigned in three distinct groups. First, there are deterministic models that transform into ensemble learners when they are given different input scenarios. Second, models that implicitly learn a posterior distribution and lastly post-process a point forecast model by including residual likelihood functions [1].

Although such PLF models can yield prediction intervals, they have rarely been paired with explainability techniques. Furthermore, new approaches are being researched based on the foundational work by Blundell et al. on Bayesian neural networks and Monte Carlo dropout. Their work demonstrates how to derive prediction intervals via weight distributions [11]. Yet integrating these uncertainty estimates with post-hoc explanations remains an open challenge in ELF.

Overall, while advances in deep learning have improved the accuracy of electricity load forecasts, especially with architectures like LSTM, they often suffer from a lack of interpretability and fail to provide reliable uncertainty estimates. Which is determined to be critical for real-world decision-making in power systems. This limitation is particularly problematic for TSOs, who must make high-stakes operational and planning decisions under growing uncertainty caused by renewable energy integration, demand variability and climate conditions.

Thus, there is a growing need for forecasting models that not only deliver high predictive performance but also offer interpretable insights and well-calibrated uncertainty estimates. Bridging this gap between accuracy, transparency and risk-awareness is the motivation of this thesis.

1.2 Research questions

The core research of this thesis engages with the question, how advanced explanibility methods and uncertainty quantification can be applied to short term electricity load forecasting.

To address this broader queston, the following subquestions are explored:

1. What are the main influence factors on the day-ahead load, and must these be preprocessed in this context?

   Method: Data engineering and data analysis.

   Expected Output: Identified key drivers and a justified preprocessing pipeline.

2. How do probabilistic forecasts from a Bayesian LSTM compare to those from standard deterministic models in terms of RMSE, MAE, and model interpretability?

   Method: Comparative model evaluation.

   Expected Output: Quantitative and qualitative comparison of forecast performance and interpretability.

3. How does embedding Bayesian uncertainty into LSTM forecasts impact the stability and fidelity of post-hoc SHAP explanations, measured via confidence intervals over SHAP values?

   Method: SHAP-based explainability analysis under uncertainty.

   Expected Output: Assessment of explanation stability and fidelity under probabilistic forecasting.

These research questions shall contribute to bridge the gap between advanced machine learning accuracy and operational transparency under uncertainty. To the authors best knowledge there is no existing research on the topic of comparing LSTM with a Bayesian LSTM architecture in terms of explainability under uncertainty in electricity load forecasting.

2 Methodology

2.1 Data Sources

The data used in this thesis comprises primarily quantitative datasets from three main sources: historical electricity load data from the ENTSO-E Transparency Platform, meteorological reanalysis data from the ERA5 (ERA5-Land hourly data from 1950 to present) [12] dataset and weather forecasts from ECMWF Open data subset [13].

The ENTSO-E data provides hourly and sub-hourly historical electricity load for Austria, reflecting aggregate national energy consumption patterns at high-voltage levels. It is structured as numerical time series data accessible programmatically through the ENTSO-E API [14]. This dataset is publicly accessible and does not require additional consent from external parties. Only the data of the Austrian bidding zone, created after the split of the former bidding zone (DE-AT-LU) in October 2018 is used.

The publicly available ERA5 dataset, obtained via the Copernicus Climate Data Store (CDS) API contains hourly weather variables such as temperature, solar radiation and wind speed. ERA5 data are provided in netCDF format, capturing spatially gridded meteorological parameters. Given the 3-month latency in the finalized ERA5 dataset, the provisional ERA5T dataset

will be employed to bridge the gap for recent data. ERA5T is updated daily but includes data up to 5-7 days prior to the current date, making it suitable for near-real-time applications but subject to later revisions. To transform these spatial datasets into usable time series for modeling, the data is spatially aggregated to produce country-level averages or specific point-based series relevant for bidding zone of Austria.

For day-ahead forecasting, additional data is needed to fill in the missing gaps for the meteorological variables. This is why ECMWF numerical weather forecasts are ingested alongside. ECMWF forecasts conviniently provide the meteorological predictions for the same variables with lead times suitable for day-ahead applications. These data are typically available in GRIB, which are similarly spatially aggregated to generate forecasting-ready time series.

2.2 Data Engineering and Preprocessing

How is the data prepared for the lstm? Prophet?

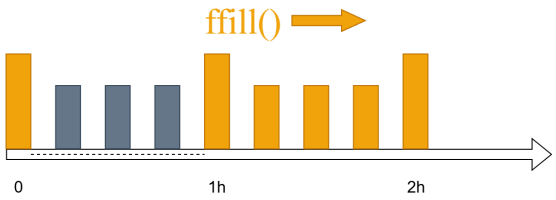

Figure 1: Weather Variables must be forward filled to match the 15 min resolution of the load timeseries

2.3 Experiment Tracking

All experiments are run locally using Jupyter Notebooks. To ensure reproducability the intermediate metrics and final models get logged to a local MLFlow-Server instance.

2.4 Model Architectures and Implementation

2.4.1 Bayesian LSTM

The implementation of a bayesian LSTM variant, relies on the former implementation BLiTZ [15] which uses a similar programatic interface as pytorch. The model architechture is shown in Figure 2.

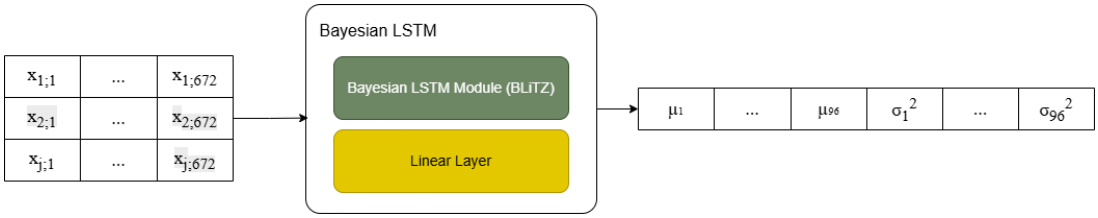

Figure 2: Input and output structure for the layesian LSTM

The model recieves a sequence of x_j input features of horizon $h = 672$ which corresponds to a lookback of 7 days based on the 15 min interval given by the structure of the load series. The output is a one dimensional vector of length two times of the forecasting horizon ($horizon_{forecasting} = 96$).

2.5 Evaluation Framework

3 Results

4 Discussion

4.0.1 Main findings

4.0.2 Comparison with previous work

4.0.3 Implications

4.0.4 Limitations

Querverweise werden in LaTeX automatisch erzeugt und verwaltet, damit sie leicht aktualisiert werden können. Hier wird zum Beispiel auf Abbildung 3 verwiesen.

Und hier ist ein Verweis auf Tabelle 1. Das gezeigte Tabellenformat ist nur ein Beispiel. Tabellen können individuell gestaltet werden.

Hier wird auf die Formel 1 verwiesen.

Figure 3: Beispiel für die Beschriftung eines Buchrückens.

Figure 4: 2. Beispiel für die Beschriftung eines Buchrückens.

Table 1: Semesterplan der Lehrveranstaltung „Angewandte Mathematik“.

Datum	Thema	Raum
20.08.2008	Graphentheorie	HS 3.13
01.10.2008	Biomathematik	HS 1.05

Table 2: 2. Semesterplan der Lehrveranstaltung „Angewandte Mathematik“.

Datum	Thema	Raum
20.08.2008	Graphentheorie	HS 3.13
01.10.2008	Biomathematik	HS 1.05

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q} \tag{1}$$

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q} \tag{2}$$

Literaturverweise sollten automatisch verwaltet werden, vor allem, wenn es viele Quellenverweise gibt. Beispiele sind [16]. Das verwendete Zitierformat (bzw. das Format des Literaturverzeichnisses) ist entspechend der Vorgaben der Studiengänge zu wählen. Es wird dringend empfohlen, Biber oder BibTeX zu verwenden (wie in diesen Beispielen).

5 Dritte Überschrift der Tiefe 1 (chapter)

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q} \tag{3}$$

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q} \tag{4}$$

Figure 5: 3. Beispiel für die Beschriftung eines Buchrückens.

Figure 6: 4. Beispiel für die Beschriftung eines Buchrückens.

Table 3: 3. Semesterplan der Lehrveranstaltung „Angewandte Mathematik“.

Datum	Thema	Raum
20.08.2008	Graphentheorie	HS 3.13
01.10.2008	Biomathematik	HS 1.05

Table 4: 4. Semesterplan der Lehrveranstaltung „Angewandte Mathematik“.

Datum	Thema	Raum
20.08.2008	Graphentheorie	HS 3.13
01.10.2008	Biomathematik	HS 1.05

The following commands are provided to provide quality of life improvements

- `\citepic` the usage of the command is depicted in figure 7.
- `\citefig` the usage of the command is depicted in figure 8.
- `\citefigm` the usage of the command is depicted in figure 9.

Figure 7: Usage of the command `\citepic`: (Source: “2017 51st Annual Conference on Information Sciences and Systems (CISS)” [16])

Figure 8: usage of the command `\citefig`: (Source: “2017 51st Annual Conference on Information Sciences and Systems (CISS)” [16], p. 30)

Figure 9: usage of the command `\citefigm`: (Source: taken with modification from “2017 51st Annual Conference on Information Sciences and Systems (CISS)” [16], p. 30)

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Documentation table of AI-based tools

AI-Based Tool	Intended Use	Prompt, Source, Page, Paragraph...
DeepL Translate	Translation of an article in English	Source (XXX), Chapter X on page X-X
Chat GPT 4.0	Grammar and Spelling	"Please list issues with spelling and grammar in the following text: ...", Entire document

List of Abbreviations

ABC Alphabet

WWW world wide web

ROFL Rolling on floor laughing

A Anhang A

B Anhang B